Experiment 22: Confidence Interval for Customer Ratings

Aim: To calculate the confidence interval for the population mean customer rating using Pandas.

Algorithm:

1. Import Pandas and the t-distribution function.
2. Store customer ratings in a DataFrame.
3. Calculate the mean and standard error.
4. Calculate the 95% confidence interval.
5. Display the mean rating and confidence interval.

Code:

Experiment 22 – Customer Ratings

```
import pandas as pd
from scipy.stats import t

df = pd.DataFrame({"rating": [4, 5, 3, 4, 5, 4, 3, 5, 4, 5]})

data = df["rating"]
n = len(data)
mean = data.mean()
se = data.std() / (n ** 0.5)
margin = t.ppf(0.975, n - 1) * se

print("Mean Rating:", round(mean, 2))
print("95% Confidence Interval:",
    (round(mean - margin, 2), round(mean + margin, 2)))
```

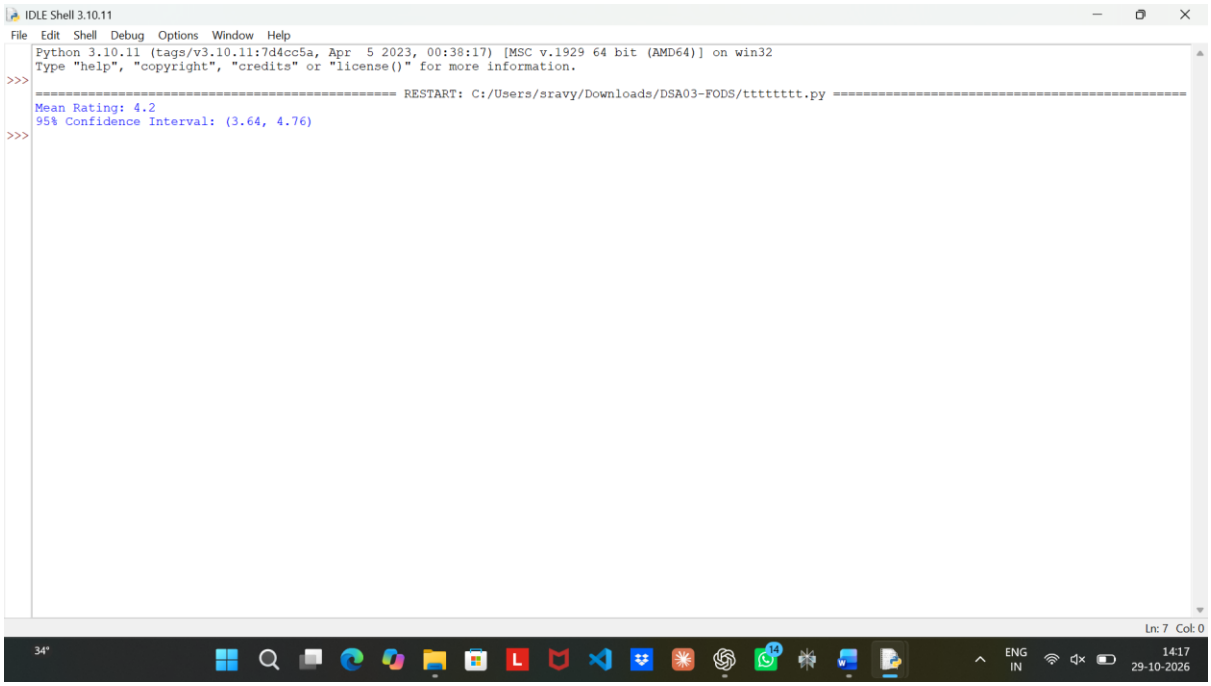
Input:

```
rating = [4, 5, 3, 4, 5, 4, 3, 5, 4, 5]
```

output:

Mean Rating: 4.2

95% Confidence Interval: (3.64, 4.76)



Result: The mean customer rating and its 95% confidence interval were calculated.